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Matching ostraca fragments using a siamese neural network

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ABSTRACT

As part of sociological studies, artifacts such as pottery ostraca from Upper Egypt are studied by egyptologists. These pottery pieces are covered with textual inscriptions many of which concern accounting and other economic and administrative matters. Thus, these writings contain pertinent information for understanding ancient Egyptian civilizations. The ostraca fragments retrieved from excavations are available in large quantities, but most of them are not yet analyzed, sorted, and reassembled by the egyptologists. We present a fragment matching approach based on pairwise local assemblies, using a 2D Siamese Neural Network to evaluate matching probabilities. This network is designed to predict simultaneously the existence or absence of a match, and the spatial relationship of one fragment in relation to the other (up, down, left, or right). We trained our deep learning model on a dataset of 6000 patches extracted from ostraca images, and achieved 96% accuracy on a validation dataset of 1000 patches. Then we propose a pipeline to reconstruct larger ostraca images using step by step pairwise matching, and able to produce a series of image reconstruction proposals. This method is based on the construction of a graph through iterative addition of small fragments. This work is intended as a proof of concept that archaeologists can benefit from automatic processing of their ostraca dataset.

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1. Introduction

Our present dataset is composed of ostraca - inscribed pottery sherds –from the site of Hut-Repit, also known as Athribis in Upper Egypt, excavated since 2005 by a German-Egyptian team led by Christian Leitz from the university of Tübingen. They date from the Graeco-Roman to the early Islamic Period (1st century BCE to 7th century CE). These sherds have mostly Egyptian inscriptions on them (most writings are in demotic, but some are also in hieroglyphs or hieratic).

Concerning two-dimensional archaeological objects reconstruction, a review published in 2015 [18] studies the evolution of methods used between 1970 and 2014, and shows that before the success of machine learning methods mostly used curve-matching algorithms based on fragment edges [16,21]. In the case of three-dimensional reconstruction [18], the same methods can be used using additional 3D information [8,15,19].

In our case, we have mostly flat fragments, and we mainly want to study the written surface of the ostraca. Thus we consider the ostraca as “documents” rather than chunks of whole objects. This lead us to focus on 2D reconstruction techniques and how to apply them in the context of ostraca images.

This work is a preliminary work to establish a proof of concept with the egyptologists working with these ostraca, first to classify sherds to group those we suppose to belong to the same ostraca or to similar ostraca from the same family, and second to propose reconstructions made of several fragments. To do this, we designed a pipeline based on a Siamese Neural Network to reconstruct ostraca from a set of artificially created fragments. Our approach is divided into two steps: local pairwise alignments using a Siamese Neural Network, and iterative global reconstruction based on a graph structure.

2. Related works

The problem of 2D image or object reconstruction from fragments has been explored in different contexts. The most common is automatic puzzle resolution. In recent years, approaches using machine learning or deep learning have shown their usefulness: for example Hammoudeh et. al. [6] used a clustering method to reconstruct images from pieces of multiple puzzles, and Paumard et al. [17] used a deep neural network model to infer relative positions of fragments from images of art pieces. A more sophisticated variant of this problem is the reconstruction of documents from

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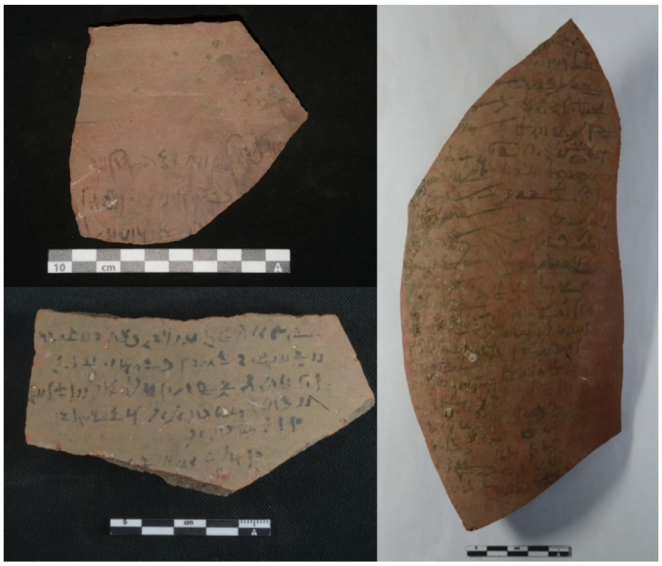

Fig. 1. Examples of ostraca fragments with scale bar below.

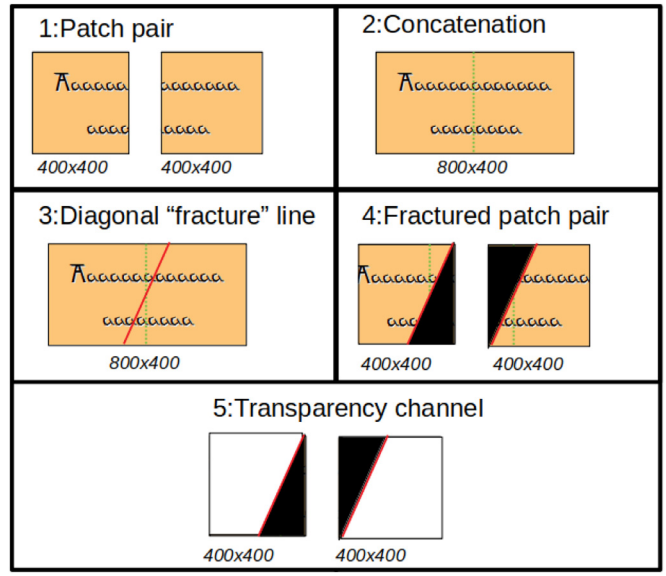

Fig. 3. Creation of artificially fractured patch pairs from regular matching pairs.

arbitrarily shredded fragments [4]. In 2019, Le et. al. [12] proposed JigsawNet, which uses a convolutional neural network for compatibility detection between pieces, followed by a global composition algorithm using graphs. da Gama Leitão et al. detail the problem of object reconstruction in the case of ancient artifacts such as potteries [5]. Such problems are edge erosion along fracture lines, partial fragments matching, and missing fragments. In this case, the information given by the shapes and contours is not sufficient to find correct alignments between pairs of fragments. However, local texture and writing patterns around the fractures are higher-level information that could be used to compute the probability of fragment matching. In this work we will propose a pipeline based on pairwise local alignments and using deep learning to reassemble image fragments step by step.

Deep Neural Networks (DNN) are multi-layer architectures designed to extract meaningful features in data, in order to solve a given task. A particular type of DNN called Convolutional Neural Networks (CNNs) such as AlexNet [11] are used with 2D or 3D inputs such as images. They consist of a sequence of convolution layers, followed by pooling layers. After the convolution layers, the computed feature-maps are fed to a classifier made of fully connected ("dense") layers. For a classification task, the CNN's outputs are the probabilities of belonging to each of the predefined class. Finally, the predicted class is the class with the highest probability.

For a task consisting of the comparison of two images, [2] designed an architecture called Siamese Neural Networks. They are made of two identical branches, that share the same weights at initialization and during all training steps. A distance or similarity function is then used to compare two inputs, and based on this the network will be trained to distinguish similar pairs from dissimilar pairs. Such networks have been used for example for face recognition [14,20], or one-shot alphabet recognition [10].

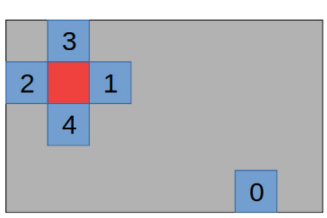

Fig. 2. Example of patch (in red) and its corresponding matching or non matching patches.

3. Local pairwise alignments

As stated previously, relying on edge complementarity only is not enough to reassemble fragments of ancient artifacts. Moreover, we noted that the approach by Paumard et. al. [17] was tested on large images representing scenes or whole objects, which enables the model to use large-scale semantic context to infer the fragments' locations. In the case of ostraca potteries, the variations of colors and inscriptions are a lot more subtle, which is why we concentrated on local pairwise matching first, and global assembly second. Thus, for fragment matching, our approach is based on a Siamese Neural Network trained to classify fragment pairs according to their hypothetical matching direction.

3.1. Dataset

The data we are using are high resolution 2D RGB images of ostraca fragments from Athribis in Egypt (see Fig. 1 for an example). The images are of different sizes, all superior to 3000 × 2000px. Since this dataset only contains one successful reassembly, we did not have ground truths for realignment tasks. To create our own ground truths, we selected 30 images of large fragments, and split them into 400 × 400px non overlapping patches. We then created matching or non matching pairs. For non matching pairs, the label 0 was affected, and for matching pairs, a label between 1 and 4 was affected according to the matching location : 1 = Right, 2 = Left, 3 = Top, 4 = Bottom (see Fig. 2).

We also randomly created "artificial fractures" between matching patches, to introduce more realistic borders (see Fig. 3). To do this, we concatenated image pairs, then drew one diagonal line passing through the alignment segment. Then using masks, we separated the two parts of the image along the diagonal. Finally, we cropped each patch to obtain our new 400 × 400px "fractured" patch pairs, centered on the "fracture" line.

To differentiate between the original image pixels and the new "fracture" pixels, we added a new channel, that we will call here "transparency". In this channel, original pixels have the value 1, and "fracture" pixels have the value 0 (see Fig. 3). To take into account illumination variations due to the pictures' acquisition, we

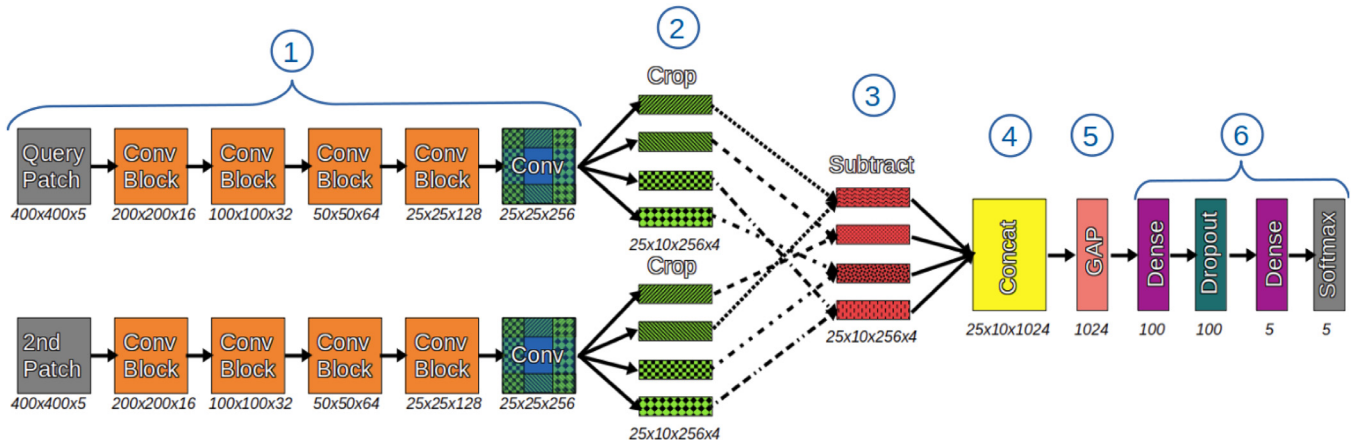

Fig. 4. Simplified architecture of our 2D Siamese network (best viewed in color). 1) Siamese CNN architecture, 2) ROI cropping in each direction, 3) Absolute value of the subtraction between ROIs from query patch to corresponding ROIs from 2nd patch (Ex: bottom of query patch / top of 2nd patch), 4) Concatenation of intermediate results for each alignment direction, 5) Global Average Pooling, 6) Classification: The output of the softmax layer is a vector of probabilities for each of the five classes.

also computed the luminance from the RGB channels, and added it as a 5th channel to our images.

3.2. Network architecture

Our proposed architecture (see Fig. 4) is a 2D Siamese convolutional network, with two branches made of a succession of "Conv blocks". Each block is made of :

- 1 2D convolutional layer
- 1 batch normalization layer [7]
- 1 leaky ReLU activation layer
- 1 2D max pooling layer

After the last convolution operation, a 10 pixel-wide band is extracted from the top, bottom, left and right of the feature-map of each Siamese branch ("Crop" layers in Fig. 4). This is done to restrict the computations to the four the four possible junction zones, in order to limit the computation time and memory usage of our neural network during training. Then "change-maps" are created by subtracting each sub feature-map from the first branch to its opposite sub feature-map from the second branch ("Subtract" layers in Fig. 4). The absolute value of these subtractions are passed to the following layer. With these two steps, we make sure that the model only considers the differences between image pairs in the four junction directions.

The change-maps are then concatenated, and fed to a Global Average Pooling (GAP) layer [13]. We used a concatenation operation to fuse the change-maps to avoid introducing asymmetry at this stage of the model. The last part of the network is made of one dense layer, one dropout layer, and one final dense layer with a softmax activation to compute the probabilities for each of the five classes. The final output is a number between 0 and 4 corresponding to one of the previously described classes (see Fig. 2).

The network was implemented using Tensorflow [1] and Keras [3] frameworks for deep learning.

3.3. Training results

We trained our network using a small dataset of 900 image patches, of which 100 were used for validation only. At each epoch training data was augmented using 90° rotations, and horizontal and vertical flips. We used cross-entropy loss function as it is a common loss used in the case of multiclass classification, and we used Adam optimizer [9].

The network converges after about 150 epochs (see Fig. 5), and the models achieves 81% accuracy on the validation set. Fig. 6 shows two examples of correctly reassembled patches, with corresponding probability superior to 0.8.

We also tested our model with a larger dataset, containing 7000 patches extracted from 160 ostraca images. However, because of memory issues, we had to downscale these patches to 200 × 200px, and remove the first "Conv block" of the network. We used 6000 patches for training, and 1000 for validation. At the

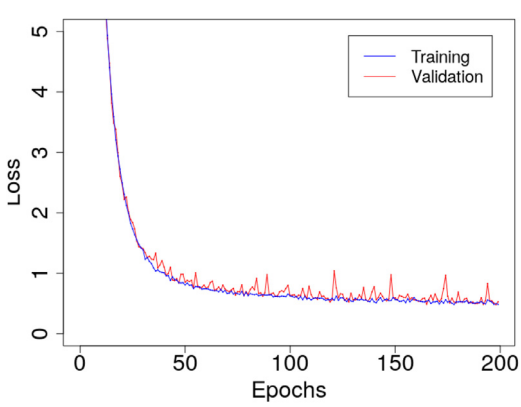

Fig. 5. Evolution of training and validation loss during training (best viewed in color).

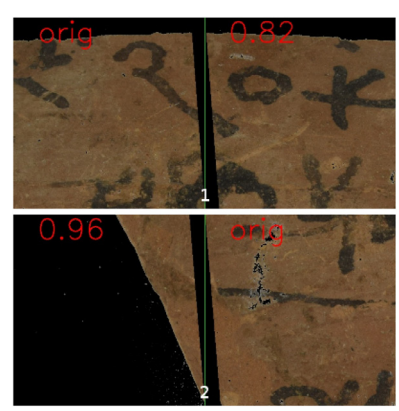

Fig. 6. Examples of two correctly reassembled patch pairs. The green line separates the two patches. "orig" patch is the query patch. For the second patch, the network's prediction probability is displayed (best viewed in color).

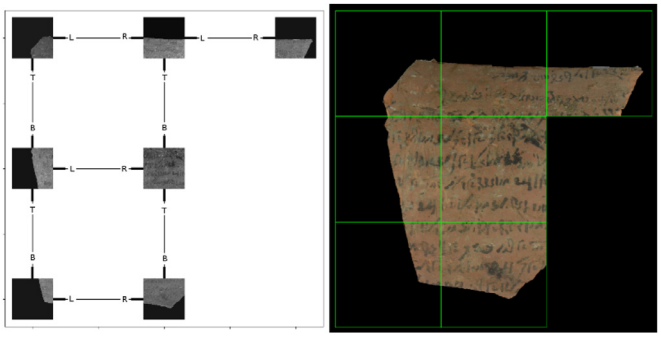

Fig. 7. Example of graph associated with assembly proposal. All edges are directed to keep track of the patches spatial relations relatively to each other.

end of training, we obtained a training accuracy of 95% and a validation accuracy of 96%.

4. Global reconstruction

Following the promising results of pairwise alignments, we wanted to go further by proposing a global reconstruction pipeline.

4.1. Methodology

Our method is based on a directed graph, dynamically constructed from successive pairwise alignments. In this graph, nodes are image patches and each of them can have up to four neighbors, one for each direction (see Fig. 7). Each directed edge between a node and its neighbor is tagged with the corresponding alignment direction, and associated probability given by the network model. The results presented in this part were obtained with the model trained on the small dataset (900 patches).

In a practical example in which we want to reassemble an image from N patches, the scenario would be the following :

1. a patch in the list becomes the “query patch” which goes in the first branch of the network
2. each of the remaining patches go sequentially through the network on the other branch
3. for each pair, the alignment direction predicted by the network is considered a good candidate if the corresponding probability is above a threshold of 0.7
4. for every new good candidate found, the current graph is duplicated and a new edge is added between the two corresponding nodes in the duplicate graph. This way, a new graph is created for each assembly proposal.
5. the same process is repeated for each patch in the list until all patches have been used as a query

In order to avoid unnecessary computations, if a node is already in a graph, only the neighbors in directions that were not found before will be accepted as good alignment candidates.

4.2. Assembly proposals

The strongly connected-components in each of the created graphs correspond to partial global alignment proposals. Fig. 8 shows examples of assembly proposals from an unknown ostraca image (Fig. 8.1). We can see in Fig. 8.2 that some patches were correctly reassembled as that the overall geometry of the ground truth ostraca was recreated.

It is apparent that even with a threshold on the pairwise alignments scores, some patches are wrongly aligned (Fig. 8.3 and Fig. 8.4). Moreover, even in a case where all pairwise alignments

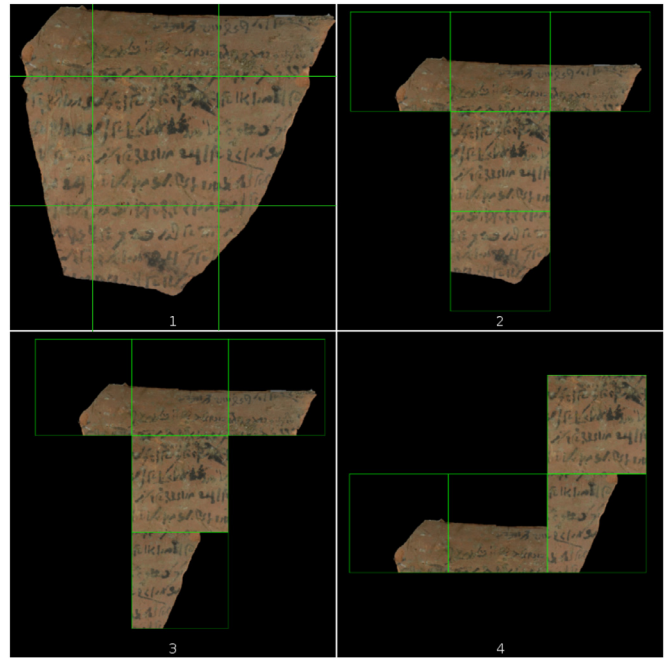

Fig. 8. Examples of partial assembly proposals generated by our graph pipeline. 1:Ground truth, 2-4:Proposed assemblies.

are correct, some fragments are missing from the global reconstruction (Fig. 8.2). These errors are due to the fact that some alignments are poorly or not at all predicted by our model. In this current state, the proposals have to be manually reviewed by a user to select the most realistic reconstructions.

During our tests, we also noted that we have poorer alignment proposals in regions without text. Our hypothesis is that the variations of texture across a single pottery are not high enough to discriminate correctly matching from non-matching fragments. Conversely, the lines created by the writings contain useful information for local matching (see in Fig. 6.1). In some ostraca, local degradation such as stains or scratches can serve a similar purpose, which helps the reconstruction.

5. Discussion

Our Siamese model is effective for pairwise assemblies because on one hand it tests if two patches are neighbors, and on the other hand it gives an information of the image geometry through the direction of alignment.

However, when we want to reconstruct a whole image from a set of patches, we obtain poor results. This is due to the fact that the correct alignment is not necessarily the one with the highest probability according to our model. Because of this, the misalignment errors add up at each step of the reconstruction. In theory, one of the graph produced by the global reconstruction pipeline should be the accurate reconstruction, but the number of possible graphs that are generated is too high to be analyzed afterwards by the user.

Another issue we encountered for this task is that the five classes that have to be predicted by our neural network are not equivalent in terms of similarity to one another. For instance, due to the fact that the text is written in horizontal lines, the two horizontal alignments (Right and Left) are harder to discriminate than the Right and Top/Bottom alignments. This is also the case between Top and Bottom alignments versus Up and Left/Right alignment. Finally, the class 0 corresponding to non-matching pairs is

easy to predict in some cases where there is an edge in one of the patches, leading to some bias during training.

To improve local alignment steps, one possibility is to add a second assembly check with another neural network which aim would be to evaluate the quality of an assembly after each pairwise alignment.

6. Conclusion

In this work, we presented a Siamese Neural Network able to take two patches of an image and to predict whether they are neighbors and if so, their position with one another. In order to train our model, we created artificial datasets of small-size ostraca fragments with simulated fractures. This model achieves 81% accuracy on our small dataset of 900 patches, and 96% accuracy on our large dataset of 7000 patches. We showed that our model is able to use texture information with small variations to predict correct pairwise alignments. We also presented a work in progress towards a global reconstruction, by creating graphs to represent possible reconstructions.

In future works, we would like to add some metadata information provided by the ceramologists and philologists working on the material, such as pottery type, script, text type,... to our model, in order to lower the risk of incorrect alignment proposals. Another application of this work would be to create an interactive interface for archaeologists, where they would see the reconstruction proposals and would be able to select the best ones or manually set thresholds to filter bad results.

Declaration of Competing Interest

Authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affi liations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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